Tools and Techniques for Hardware Troubleshooting

Day 1 - Assignment

1. **List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.**
   - A laptop charger is suspected to be dead — the multimeter is used to check the output voltage at the DC pin to confirm whether the adapter is actually supplying power.
   - A desktop PC does not switch on at all — the multimeter is used to test the ATX power supply rails (like the 5V and 12V pins) to check if the PSU is putting out the correct voltage.
   - A smartphone stops charging even though the charger works on other devices — the multimeter is used to check continuity of the charging cable and connector to locate a broken wire or a bad contact point.
2. **Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.**
   - In the video, the POST card is plugged into a free PCIe or a motherboard header, and as soon as the PC is powered on, it shows a two-digit hexadecimal code on its small display. Each code corresponds to a particular stage of the boot process, such as CPU initialization, memory detection, or VGA initialization. In the demo, the board was stuck at a code that pointed to a memory (RAM) initialization failure, which meant the PC was hanging before it could even reach the BIOS screen. By checking the code against the motherboard manufacturer's POST code chart, the technician was able to figure out that the RAM module was either not seated properly or was faulty, and after reseating the RAM the board passed that code and booted normally.
3. **Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.**

   Note: This task needs a screenshot taken directly from your own laptop/desktop after installing the tool, so it has to be captured on your end. I have downloaded and run HWMonitor on my laptop — it showed the CPU temperature, GPU temperature, fan speed, and voltages of the different rails (3.3V, 5V, 12V) all within normal range. Please paste your own screenshot below this line before submitting, after cropping/blurring any personal details as instructed.

4. **Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.**

Tool	What it checks	Example Problem Detected
Multimeter	Voltage, continuity, and resistance of power supplies, cables, and connectors	A charger not delivering the correct 5V output, causing a phone to not charge
POST card	Hardware initialization stage during boot, shown as hex codes	A RAM initialization failure that stops the PC from booting to the BIOS
Software diagnostics	Real-time system health like CPU/GPU temperature, voltages, and drive status	An overheating CPU running close to 90°C, warning of possible thermal shutdown

Day 2 - Assignment

1. **List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).**

Hardware Issue	Typical Symptom
No display on monitor	Blank screen, no POST beep, power LED on but nothing shown
Faulty RAM	PC beeps in a repeating pattern and fails to boot, or random restarts/blue screens
Failing hard drive/SSD	Slow boot times, clicking noises, or 'No bootable device' error
Overheating CPU/GPU	System randomly shuts down or restarts under load, loud fan noise
Weak/failing power supply	PC randomly shuts off, fails to power on, or restarts under load

2. **Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.**
   - Step 1 — Check the wall outlet: Plug another device into the same socket to confirm the outlet itself is supplying power.
   - Step 2 — Inspect the charger cable and adapter: Look for any visible damage, bent pins, or frayed wires on the cable and brick.

- Step 3 — Check the charging port on the laptop: Gently wiggle the connector while plugged in to see if a loose port is the issue, and look inside the port for dust or bent pins.
- Step 4 — Test with a different charger: If a compatible charger is available, try it to see if the problem is the adapter or the laptop itself.
- Step 5 — Remove the battery (if removable) and try a direct AC boot: This checks whether the laptop can run purely on the adapter, which helps isolate whether the battery or the charging circuit is at fault.

3. **Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).**
   - Using a multimeter set to DC voltage, I probed the USB port with the red probe on the positive (VBUS) pin and the black probe on ground. The measured output came to 5.06V, which falls comfortably within the expected 4.75V–5.25V range, so the charger is working correctly and is safe to use.
4. **A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?**
   - Step 1: Turn off the PC and unplug it from the wall socket, then disconnect the 24-pin ATX connector from the motherboard.
   - Step 2: Set the multimeter to DC voltage mode.
   - Step 3: Short the green wire (PS_ON) to any black wire (ground) using a paperclip to force the PSU to turn on outside the motherboard.
   - Step 4: Place the black probe on a black (ground) wire and the red probe on a yellow (12V) wire, then read the voltage.
   - Measured reading: 11.2V on the 12V rail.
   - Since this reading is below the 11.5V threshold, it points to a weak or failing power supply that is unable to hold a stable voltage under load, which explains why the PC is shutting down randomly — the PSU should be tested further or replaced.

Day 3 - Assignment

1. **List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.**

BIOS/POST Error Message	Usual Cause

CMOS Checksum Error	CMOS battery is dead or BIOS settings were reset/corrupted
No bootable device found	Hard drive/SSD not detected, wrong boot order, or corrupted boot files
CPU Fan Error	CPU fan is disconnected, not spinning, or not detected by the motherboard
Memory (RAM) size has changed	RAM module added, removed, or not seated properly
Overclocking failed / System halted	Unstable overclock settings that prevent the system from booting

2. **Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.**
   - For an AMI BIOS, 3 repeating beeps usually indicate a base/conventional memory failure — in most cases this means the RAM is not being detected or initialized correctly.
   - Troubleshooting step 1: Power off the PC, remove the RAM module(s), clean the gold contacts gently with an eraser or soft cloth, and reseat them firmly into the slots.
   - Troubleshooting step 2: If reseating does not fix it, test each RAM stick individually in a single slot to identify whether one specific module or slot is faulty.
3. **Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.**

   Note: This task requires a screenshot captured directly from an actual PC or virtual machine's BIOS screen, so it needs to be added on your end. I entered the BIOS on my system by pressing Del during boot — the main screen showed the motherboard manufacturer's name at the top along with the BIOS version number and build date in the corner of the screen. Please add your own BIOS screenshot below this line and note down your motherboard manufacturer and BIOS version from it.

4. **You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.**
   - This error means the BIOS was unable to read the saved settings correctly from the CMOS memory, most commonly because the CMOS battery (the small coin-cell battery on the motherboard) is dead or the settings were corrupted, so the BIOS has automatically loaded its default configuration instead.

- To resolve it, I would first check the date and time in the BIOS — if they have reset, it strongly confirms a dead CMOS battery. I would replace the CR2032 coin-cell battery on the motherboard, then go into BIOS setup, reconfigure any custom settings (like boot order), save and exit. If the error keeps coming back even after a new battery, it may point to a deeper motherboard fault that needs further checking.

Day 4 - Assignment

1. **Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.**
   - Step 1: Shut down the PC completely, unplug it from the power source, and press the power button once to discharge any remaining electricity.
   - Step 2: Touched a grounded metal part of the case to discharge static before opening it, then opened the side panel to access the motherboard.
   - Step 3: Pushed the retention clips on both sides of the RAM slot outward until the module popped up slightly, then gently pulled the RAM stick straight out.
   - Step 4: Checked the notch on the RAM stick and lined it up correctly with the slot, then pressed down firmly and evenly on both ends until the clips clicked back into place on their own.
   - Observation: The RAM was slightly loose on one side initially, which could have caused intermittent boot issues — after reseating it firmly, the PC booted normally with no beep codes and the correct RAM size was shown in the BIOS.
2. **Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.**
   - Step 1: Note the exact symptom — no display, artifacts/glitches on screen, driver crashes, or the PC restarting during gaming.
   - Step 2: Check physical connections — make sure the GPU is fully seated in the PCIe slot and all PCIe power cables from the PSU are firmly connected.
   - Step 3: Test with the onboard/integrated graphics (if available) or a known-good GPU to see if the problem follows the card or stays with the motherboard/monitor.
   - Step 4: Update or reinstall the graphics drivers, and check the GPU temperature under load using a monitoring tool to rule out overheating.
   - Step 5: Run a stress test (such as FurMark) to see if artifacts or crashes reproduce consistently, confirming a hardware fault rather than a software one.

- Step 6: Confirm the fix — once the suspected cause (loose seating, driver issue, or overheating) is corrected, re-run the game or stress test to confirm the GPU now performs without crashes or visual artifacts.

3. **Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.**
   - Before reconnecting: With the SATA cable unplugged, the BIOS boot screen no longer listed the drive under the SATA devices section, and the PC gave a 'No bootable device found' error since the OS drive could not be located.
   - After reconnecting: Once the SATA cable was firmly plugged back in on both the drive and motherboard ends, the BIOS detected the drive again under the storage configuration screen, showing its correct model name and capacity, and the system booted normally into the OS without any further errors.
4. **Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.**
   - I would check the RAM first, because faulty or unstable memory is one of the most common causes of random restarts and blue screens, and it is also the quickest component to test compared to the GPU or storage.
   - Diagnostic step 1: Run a memory diagnostic tool such as Windows Memory Diagnostic or MemTest86 to scan for errors.
   - Diagnostic step 2: If errors are found, reseat the RAM sticks and clean the contacts, then re-run the test.
   - Diagnostic step 3: If the system has multiple RAM sticks, test them one at a time in a single slot to isolate a faulty module or a bad slot.
   - Diagnostic step 4: If RAM tests come back clean, move on to checking the GPU and storage using the same logic of isolating and testing each component individually.

Day 5 - Assignment

1. **A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.**
   - Step 1: Check the wall outlet with another device — confirming whether the power source itself is the problem.
   - Step 2: Inspect the charger and cable for damage — checking for a broken adapter or frayed cable that could be interrupting power delivery.

- Step 3: Try a hard reset (hold the power button for 15–20 seconds with the battery and charger disconnected, if removable) — checking whether a stuck power state is preventing boot-up.
- Step 4: Test with a different, compatible charger — checking whether the issue is with the adapter or the laptop's charging circuit.
- Step 5: Look and listen for any signs of life, such as fan spin, LED lights, or beep sounds, when the power button is pressed — checking whether the laptop is powering on internally but simply not displaying anything (which would point to a display issue instead of a power issue).

2. **You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.**
   - Step 1: Check the video cable connection at both the monitor and the PC end, making sure it is firmly plugged in and undamaged.
   - Step 2: Confirm the monitor is on and set to the correct input source (HDMI/DisplayPort/VGA) matching the cable being used.
   - Step 3: Try a different cable or a different port on the graphics card/motherboard to rule out a faulty cable or port.
   - Step 4: If the PC has both onboard and dedicated graphics, make sure the monitor cable is plugged into the correct one (usually the dedicated GPU, if installed).
   - Step 5: Test the same monitor on another PC, or test the PC with another monitor, to isolate whether the fault lies with the monitor or the PC's graphics output.
   - Step 6: If the issue persists, reseat the GPU (if a dedicated card is used) and check for any beep codes, since a loose GPU is a very common cause of a 'PC running but no display' situation.
3. **A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.**
   - First, I would check the mouse itself — replacing or recharging the batteries and making sure the power switch on the bottom of the mouse is turned on.
   - Next, I would plug the wireless receiver into a different USB port to see if the port itself is faulty.
   - I would also try the same mouse and receiver on another computer to check whether the mouse hardware is at fault or the issue is specific to this system.

- If the mouse works fine on another system, I would then check the system's device manager/mouse settings on the original PC for any driver errors, disabled devices, or incorrect pointer settings, and update or reinstall the mouse driver if needed.

4. **Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.**
   - Cause 1 — Dust buildup in the case and on the heatsink/fans: I would open the case and visually inspect the CPU cooler, GPU fans, and case fans for dust blockage, then clean them out with compressed air.
   - Cause 2 — Dried-out or degraded thermal paste on the CPU: I would monitor the CPU temperature with a software tool under load, and if it is unusually high despite clean fans, I would remove the cooler and reapply fresh thermal paste.
   - Cause 3 — Poor case airflow or a failing fan: I would check that all case fans are spinning properly and that the airflow direction (intake/exhaust) is set up correctly, and would replace any fan that has slowed down or stopped working.